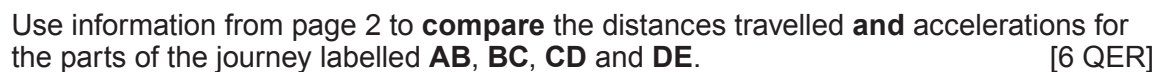


- 3430UF01
07



- (b) The diagram shows a 5-carriage bi-modal electric/diesel train.



The table below shows information about two types of bi-modal electric/diesel trains.

Train	Mass ($\times 10^5$ kg)	Maximum speed (m/s)	Standard acceleration (m/s^2)	Standard deceleration (m/s^2)	Emergency deceleration (m/s^2)
5 carriage	2.3	55.8	0.7	1.0	1.2
9 carriage	4.4	55.8	0.7	1.0	1.2

For the journey from Swansea to London **two** of the 5-carriage trains are joined, making a 10-carriage train.
The 10-carriage train has the same speed and acceleration as a 5-carriage train.

- (i) State Newton's second law of motion as an equation. [1]

- (ii) Use information from the table to answer the following questions.

- I. Use an equation from page 2 to calculate the resultant force needed to accelerate the 10-carriage train. [2]

Resultant force = N

- II. Use the equation:

$$\text{acceleration} = \frac{\text{change in velocity}}{\text{time}}$$

to calculate the time taken to accelerate the 10-carriage train from rest to its maximum speed. [2]

Time = s



- (iii) A student states that the 9-carriage train is approximately **double** the mass of the 5-carriage train and if they are both travelling at the same speed it will take approximately **double** the time to stop in an emergency. Explain whether the **2 claims** made by the student are correct. [3]

.....

.....

.....

.....

.....

